4 April 1955
SUPERSEDING
MIL-M-13574 (Ord)
9 August 1954

MILITARY SPECIFICATION
MOTOR, INDUCTION, A-C, 208 VOLT, 1-1/2 H.P., 3 PHASE, 60 CYCLE

1. SCOPE

1.1 This specification covers a motor for use as a component of the Gear, Oil, M3A1 with Carriage, Gun, 40mm, AA, M2A3.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps

F8202351	Motor, Induction

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements specified in Specification MIL-F-13926 which are common to all fire control instruments shall be applicable to this specification.

3.2 Material.- Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

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3.3 Electrical

3.3.1 Balance.- The rotor shall be in dynamic balance at all speeds up to and including 8000 rpm. The balancing shall be accomplished by such means as will preclude the possibility of balancing material shifting and destroying rotor balance. Solder shall not be used as a method of balance. The rotor shall be mounted in ball bearings of the best grade.

3.3.2 Rating.- The motor shall have a continuous duty rating of 1-1/2 h.p. at rated input voltage.

3.3.3 Speed.- The speed of the motor at rated load shall be not less than 3500 rpm when the motor is operating within an ambient temperature range from 60°F to 90°F.

3.3.4 Developed horsepower.- The motor shall be capable of developing 3 h.p. for 15 seconds.

3.3.5 Locked rotor torque.- Locked rotor torque shall be not less than 200 percent nor more than 250 percent rated load torque.

3.3.6 Frequency variation.- The motor shall operate satisfactorily at rated load and rated voltage with a frequency variation of ± 5 percent of the nameplate rating.

3.3.7 Voltage variation.- The motor shall operate satisfactorily at rated load and rated frequency with a voltage variation of 10 percent of the nameplate rating.

3.3.8 Voltage and frequency variation.- The motor shall operate satisfactorily at rated load with a combined variation in voltage and frequency of ± 10 percent of the nameplate rating.

3.3.9 High potential.- There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength of the insulation sufficient to cause failure of the motor when tested in accordance with 4.2.1.

3.3.10 Insulation resistance.- The insulation resistance shall be not less than 500 megohms when tested in accordance with 4.2.2. The insulation resistance on motors subjected to the temperature rise test, shall be not less than 50 megohms when measured immediately following the temperature rise test. The test shall be conducted in accordance with 4.2.2.

3.3.11 Temperature rise.- The temperature rise of the motor shall not exceed 60°C for continuous duty at rated load when tested in accordance with 4.2.3.

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3.4 Environmental

3.4.1 Temperature.- The motor shall meet the requirements specified in 3.3.2 and 3.3.3 throughout the air temperature range extending from plus 51.5°C (+ 125°F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr.) to minus 40°C (- 40°F) (minimum exposure of 3 days without benefit of solar radiation). The motor shall be capable of safe storage throughout the air temperature range extending from plus 71°C (+ 160°F) (for periods as long as 4 hours per day) to minus 62.2°C (- 80°F) (for periods of at least 3 days duration). Test shall be in accordance with 4.3.2.

3.4.2 Humidity.- The motor shall not suffer damage or fail to meet the requirements specified in 3.3.2 and 3.3.3 immediately after being subjected to the humidity test specified in 4.3.3.

3.4.3 Rain.- The motor shall show no evidence of water penetration and shall not suffer damage immediately after being subjected to the rain test specified in 4.3.4.

3.4.4 Vibration.- The motor shall not suffer damage or fail to meet the requirements specified in 3.3.2 and 3.3.3 immediately after being subjected to the vibration test specified in 4.3.5.

3.4.5 Shock.- The motor shall not suffer damage or fail to meet the requirements specified in 3.3.2 and 3.3.3 immediately after being subjected to the shock test specified in 4.3.6.

3.5 Workmanship.- Workmanship shall be in accordance with Specification MIL-F-13926.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified each motor shall be inspected for conformance with this specification and with Specification MIL-F-13926. Unless otherwise specified, tests shall be performed at an ambient temperature between 60°F and 90°F, a barometric pressure between 28 and 32 inches of mercury, and a relative humidity not greater than 80 percent.

4.2 Electrical

4.2.1 High potential.- The insulation between ungrounded circuits and ground, and between isolated circuits shall be subjected to a potential of 1,000 volts (rms) plus twice the rated voltage at a frequency of 60 cycles per second. One of the first five and one out of every additional 50 or fraction thereof shall be tested. The test voltage shall be applied at a rate of 500 volts (rms) per second until the test potential is reached. The test potential shall be maintained

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for one minute. For all other units the test voltage shall be applied instantaneously for a period not less than 5 seconds or more than 30 seconds. The unit shall meet the requirements specified in 3.3.9. Insulation resistance test shall be performed immediately after this test.

4.2.2 Insulation resistance.- The insulation resistance between ungrounded circuits and ground, and between isolated circuits shall be measured by an approved method using a source of d-c potential of 500 volts. The unit shall meet the requirements specified in 3.3.10.

4.2.3 Temperature rise.- Three of the first five motors and one out of every additional 50 or fraction thereof shall be tested. The motor shall be tested in still air and shall not be mounted in a manner that will cause heat dissipation through the mounting. The motor shall meet the requirements specified in 3.3.11. Immediately after this test the insulation resistance test shall be performed. The temperature rise test shall be determined as follows:

(a) A stop watch shall be started at the instant the unit is stopped.

(b) Necessary bridge connections shall be made.

(c) A value of resistance which is slightly smaller than the resistance of the winding shall be preset into the bridge. Observe the null indicator of the bridge; and the instant the needle crosses the "zero" position, record the time and the value of the resistance set into the bridge. This procedure shall be followed until five values of resistance versus time after shutdown are recorded. The resistance values shall then be plotted against the corresponding time, and the curve extrapolated to zero time. The value of (R) at zero time shall be used in equation (2) in order to obtain "T". The value"T" shall then be used in equation (1) to obtain "T_r".

(1) $T_r = T - T_a$

(2) $T = \frac{R}{r} (234.5 + t) - 234.5$

Where: T_r = Temperature rise in degrees centigrade.

T = Winding temperature in degrees centigrade.

T_a = Ambient temperature in degrees centigrade at end of temperature rise test.

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R = Hot resistance of winding in ohms.

r = Cold resistance of winding in ohms.

t = Temperature of winding in degrees centigrade when cold resistance measurement was made.

4.3 Environmental

4.3.1 General.- The environmental tests specified shall be limited to those contained in the following subparagraphs. Three of the first five motors and one out of every additional 50 or fraction thereof shall be tested.

4.3.2 Temperature.- The motor shall be exposed to an air temperature of minus 62.2°C (-80°F) for a period of 3 days. The air temperature shall then be raised to minus 40°C (- 40°F) and after the internal temperature of the motor has reached equilibrium (approximately 4 hours), it shall be tested for conformance with 3.4.1. Raise the air temperature to plus 71°C (+160°F) and after the internal temperature of the motor has reached equilibrium at that temperature allow it to remain at that temperature for a period of 4 hours. The air temperature shall be reduced to plus 51.5°C (+ 125°F) and after the internal temperature of the motor has reached equilibrium at that temperature expose the top surface to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The motor shall then be tested for conformance with 3.4.1.

4.3.3 Humidity.- A test chamber that is capable of being sealed shall be used; the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the unit being tested. A motor shall be placed in the chamber, the temperature of which is adjusted to plus 46.1°C ± 1.1°C (+ 115°F ± 2°F) and the relative humidity adjusted to 95 ± 5 percent. The unit shall remain under the above conditions for a period of 6 hours. The temperature of the chamber shall be adjusted to plus 26.7°C ± 1.1°C (+ 80°F ± 2°F) and the motor shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated 10 times. Immediately after this test the unit shall be tested for conformance with 3.4.2.

4.3.4 Rain.- The motor shall be mounted in the test chamber to simulate installation conditions. The rain test chamber temperature shall be maintained between 20°C and 30°C (68°F and 86°F) throughout the test period. A simulated rainfall of 4 + 1 inch per hour shall be produced by a suitable means of such design that the water is emitted in the form of small droplets rather than a fine mist. The temperature of the water shall be maintained between 11°C and 24°C (51.8°F and 75.2°F). The water spray shall be directed on the motor

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for a period of 2 hours in directions ranging from vertical to 45 degrees from the vertical. The simulated rain shall be directed on all surfaces of the motor that are normally subjected to rain when the unit is in its operating position. Immediately after this test the unit shall be tested for conformance with 3.4.3.

4.3.5 Vibration.- The motor shall be mounted rigidly using its own mounting provisions to simulate field use and shall be vibrated for 30 minutes along each of the three mutually perpendicular directions. Each unit tested shall be subjected to a simple harmonic motion having an amplitude of 0.031 inch (0.062 inch total excursion) at 55 cycles per second. The entire range of frequencies from 10 to 55 cycles per second and return to 10 cycles per second shall be traversed in one minute. Immediately after this test the unit shall be tested for conformance with 3.4.4.

4.3.6 Shock.- The motor shall be mounted rigidly, using its own mounting provisions, to simulate field use. A shock of 20 g's shall be imparted three times in each direction along each of the three mutually perpendicular directions making a total of 18 shock impulses. Each shock impulse shall have a time duration of 11 ± 1 milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental from the measured shock impulse. Immediately after this test the unit shall be tested for conformance with 3.4.5.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Level of packaging, packing and marking shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data.- Contracts and invitations for bids should specify the title, number, date of this specification and level of packaging and packing.

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